

Problem 1.1 (a) $1\text{ps} = 10^{-12}\text{s}$, $1\text{fs} = 10^{-15}\text{s}$;

(b) $1\text{nm} = 10^{-9}\text{m}$;

(c) $1\text{MeV} = 10^6\text{eV}$, $1\text{GeV} = 10^9\text{eV}$

Problem 1.3 The mass of the earth

$$m \approx 10\text{mol} \times 6.022 \times 10^{23} / \text{mol} \times 1\text{kg}$$
$$= 6.022 \times 10^{24} \text{kg}$$

$$\text{accuracy} = \frac{6.022 \times 10^{24} - 5.98 \times 10^{24}}{5.98 \times 10^{24}} = 0.702\%$$

Problem 1.4 the number of seconds in a year

$$\frac{\pi \times 10^7 - 365.24 \times 86400}{365.24 \times 86400} = 0.446\%$$

Problem 1.5 logarithmic scale

$$\lg x = \frac{1}{2} (\lg 10^{-15} + \lg 10^{26}) = 5.5;$$

$$x = 10^{5.5} = 3.16 \times 10^5;$$

$$\ln x = \frac{1}{2} (\ln 10^{-15} + \ln 10^{26}) = 5.5 \ln 10 \quad \text{same}$$

Problem 1.8

$$\left(\frac{Gm}{R}\right)^{1/2}$$

One way:

$$G^{\alpha} m^{\beta} R^{\chi}$$

Alternatively: similar to $v^2 = 2as$

$$v^2 \sim gR \sim G \frac{m}{R^2} R$$

Problem 1.9 (optional) Similar to gravitational self-energy, electrostatic self-energy is

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

Set

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r} = m_e c^2$$

and we have

$$r = \frac{1}{4\pi\epsilon_0} \frac{e^2}{m_e c^2} = 2.82 \times 10^{-15} \text{ m}$$

(see section 15.5)

Incorporating some *universal constant(s)* in we might get some results beyond the basic requirement: G , h , $\hbar$ etc.

Scheme 2

$$G \frac{m_e^2}{r} = m_e c^2$$

$$r = \frac{Gm_e}{c^2}$$

Gravitational radius (4.2.13):

$$r = \frac{2Gm}{c^2}$$

Scheme 3 Find a wavelength $\lambda \sim r$

A wavelength may corresponds to an energy

$$h\nu = \frac{hc}{\lambda}$$

It may correlate to energy of a microscopic particle. Set

$$\frac{hc}{\lambda} = m_e c^2$$

We have Compton wavelength Eq.(22.3.3):

$$\lambda = \frac{h}{m_e c}$$

Other possibilities

$$a_0 = 4\pi\epsilon_0 \frac{\hbar^2}{m_e e^2}$$

$$R_\infty^{-1} = \frac{64\epsilon_0^2 \pi^3 \hbar^3 c}{m_e e^4}$$

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Problem 1.12

Solution: At extreme low temperature,

$$C_V = \gamma T$$

Problem 1.14

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

polar vector

Problem 1.17

$$b_i \cdot a_j = 2\pi \delta_{ij}$$

$$\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

(triple products in denominators are cyclic)

$$b_1 = 2\pi \frac{a_2 \times a_3}{a_1 \cdot (a_2 \times a_3)}, b_2 = 2\pi \frac{a_3 \times a_1}{a_1 \cdot (a_2 \times a_3)},$$

$$b_3 = 2\pi \frac{a_1 \times a_2}{a_1 \cdot (a_2 \times a_3)}$$

Problem 1.19(optional)

$$m\boldsymbol{\omega} \times \mathbf{v} = q\mathbf{v} \times \mathbf{B} = -q\mathbf{B} \times \mathbf{v}$$

We may set

$$\left(\boldsymbol{\omega} + \frac{q}{m} \mathbf{B} \right) \times \mathbf{v} = 0$$

While $\mathbf{B}$ is known, $\mathbf{v}$ is arbitrary. So

$$\boldsymbol{\omega} = -\frac{q}{m} \mathbf{B} \qquad \boldsymbol{\omega} + \frac{q}{m} \mathbf{B} = \lambda \mathbf{v}$$

See Eq.(9.5.5) for details